

AI Voice Screening Platform

Executive Overview: Technology Stack, Hosting Architecture & Cost Model

TARGET BUDGET: < \$60/MO

DEVELOPMENT COST: \$0.00

REGION: SRI LANKA / SINGAPORE

1. Executive Summary & Total Monthly Cost

By connecting your existing Next.js web application with pay-as-you-go speech APIs and open-source orchestration, variable AI costs are minimized while eliminating fixed monthly subscription fees.

ESTIMATED TOTAL MONTHLY OPERATING COST

~\$35.50 / Month

Based on 1,500 Call Minutes per Month (~50 Screening Calls / Day)

2. Complete Technology Stack

Component	Provider / Technology	Function in System	Pricing Rate	Est. Monthly Cost
Telephony Carrier	Telnyx SIP	Connects phone calls to Sri Lankan mobile networks (+94)	~\$0.007 / min	~\$10.50
Speech-to-Text (STT)	Deepgram Nova-2	Converts live candidate voice responses into text	~\$0.005 / min	~\$7.50
AI Brain (LLM)	OpenRouter (gpt-4o-mini)	Interviews candidate, evaluates answers, detects language	~\$0.001 / min	~\$1.50
Text-to-Speech (TTS)	Cartesia Sonic	Synthesizes natural, human sounding interviewer voice	~\$0.010 / min	~\$15.00
Virtual Phone Number	Telnyx DID	Dedicated calling number rental	Flat rate	~\$1.00
Voice Framework	Pipecat.ai	Open-source real-time audio streaming engine	Open Source	\$0.00
Database & Control Panel	Existing Next.js Stack	Stores candidate logs, transcripts, and screening status	Existing Infra	\$0.00
Server Hosting	Existing Server Infra	Background worker process for audio streams	Existing Infra	\$0.00

3. Hosting & Deployment Architecture

1. Next.js Web Application (Existing Infrastructure)

Acts as the control plane and dashboard. It triggers outbound calls via API requests to Telnyx and receives finalized call transcripts, salary expectations, and notice periods directly into your existing database.

2. Pipecat Python Audio Engine (Background Worker)

Runs a lightweight WebSocket server that holds live audio streams open with Telnyx. It can be hosted on your existing server setup or deployed to Render / Railway.

Crucial Latency Rule for Sri Lanka: Deploy your audio worker explicitly in the **Singapore Region (ap-southeast-1)**. Connecting Sri Lankan telecom providers (Dialog, SLT-Mobitel, Airtel, Hutch) to Singapore keeps audio latency under 40ms, ensuring natural conversation flow without voice delay.

4. \$0.00 Free Testing Strategy

You can build, integrate, and run hundreds of real test calls completely for **\$0.00** before spending any money:

SERVICE PROVIDER	FREE SIGNUP CREDIT / TIER	TESTING CAPACITY INCLUDED
Deepgram (STT)	\$200.00 Free Credit (No Credit Card)	~40,000 minutes of free transcription
Cartesia (TTS)	20,000 Free Monthly Credits	~130+ minutes of voice generation
Telnyx (Telephony)	\$5.00 Free Testing Credit	~700 minutes of real mobile phone calls
OpenRouter (LLM)	Free LLM Models (e.g., llama-3.2:free)	100% free test reasoning and conversation